

Scooby-Dooby-Data

Modeling to Predict Ratings



Lily Bertlshofer, Maven Brown,
Levin Wu, and Bridget Baker

Our Question:

Can we predict the IMDB rating of Scooby Doo episodes? What factors affect episodes IMDB ratings the most? Does this change over time?



Why This is Interesting

01

Ratings are important for shows to obtain their budget

Identifies factors which affect ratings

02

03

Cultural changes can be seen in what factors affect an episode over time.

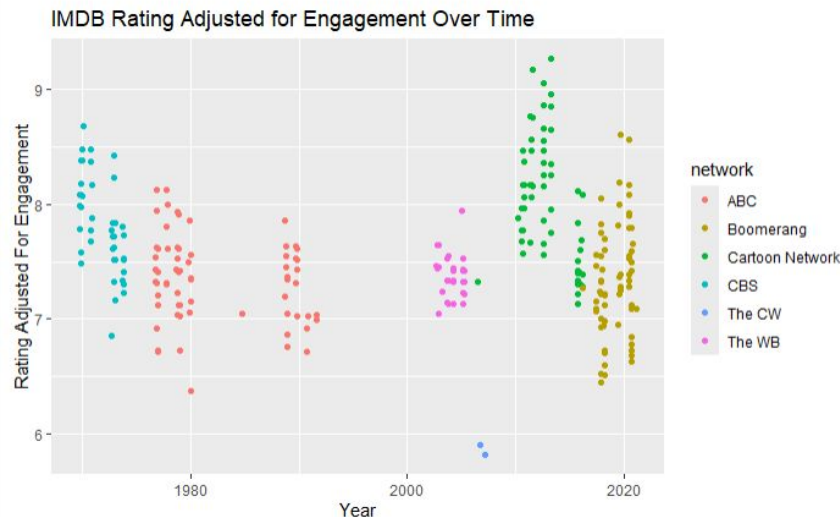
The Data

SOURCE	William C. Schooleman (plummye on kaggle)
OBSERVATIONS	615, each a unique Scooby Doo media
VARIABLES	75, many different data types
Characters	Series name, title, monster name/type/species, motive, air date
Factors	Network, season, format
Numeric	Rating, engagement, runtime, monster/culprit/suspect count
Boolean	caught/captured/unmask/snack for each member, suspect arrested

Data cleaning

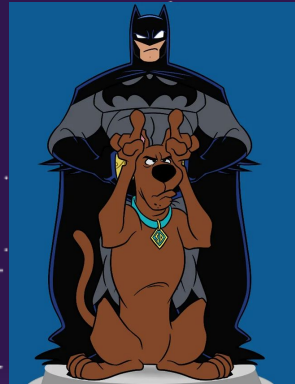
There were several issues in working with the data:

- Some of the variables had too many factors to use a cross validation technique, so they had to be removed from the data in order to make the model.
- Some episodes were not framed like a normal Scooby Doo episode, like the series *Laff-A-Lympics*
- We adjusted the IMDB rating variable to account for the engagement



The IMPORTANT Variables

- Network: what network the episode originally aired on
- Caught.*character*: which character caught the villain
- Scrappy.doo: Was Scrappy Doo in the episode
- Monster.real: Was the monster a real monster
- Batman: was batman in the episode
- Captured.*character*: Was this character captured by the villain during the episode
- Scooby.dum: was Scooby Dum in the episode



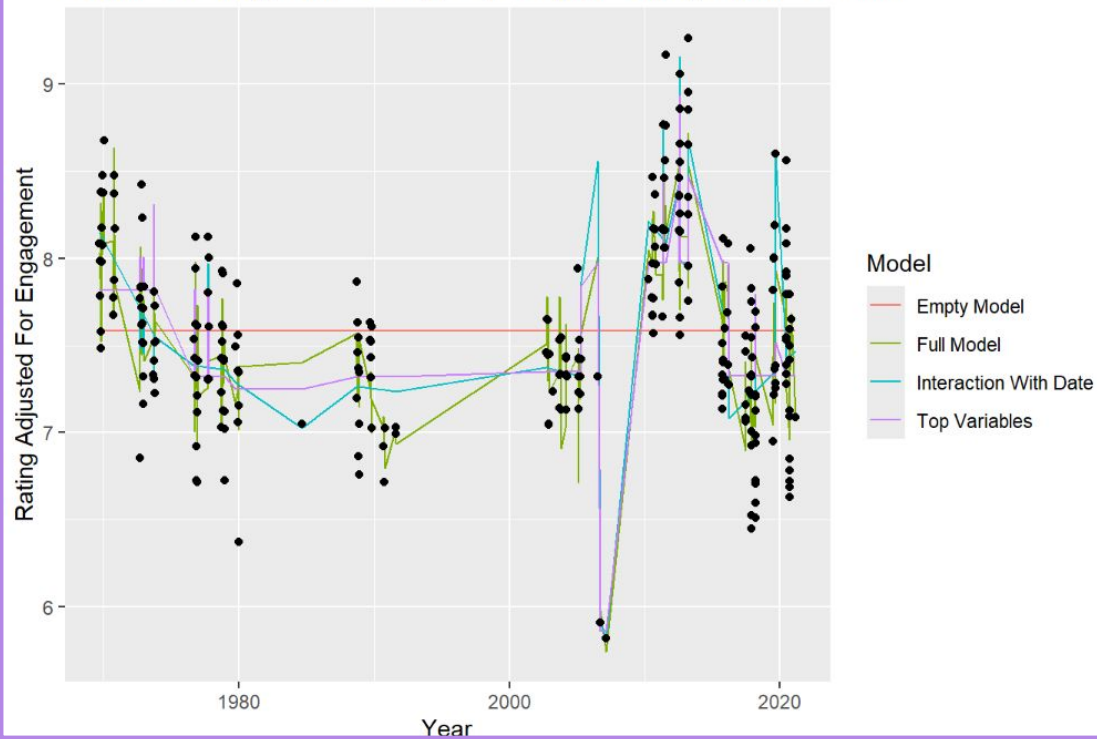
The Models

- 01 Full Model: predicts rating based on all other variables
- 02 Top Variables: predicts rating based on selection of strongest variables from the full model (network + scrappy.doo + snack.scooby + batman + caught.not + monster.real + scooby.dum)
- 03 Interaction With Date: Top variables along with interaction of date aired
- 04 Empty model: empty model, just the mean IMDB score
- 05 Full Model Pre/post 1995: split the dataset into two sets from before and after midpoint of the show (1995)

	model	MAE
1	Full model	0.216688281213974
2	Empty Model	0.372172294149406
3	Top Variables	0.289571373518429
4	Interaction w/ Date	0.250314701202146
5	Full model pre-1995	0.298534709160051
6	Full model post-1995	0.31254563485408

Results

Models Showing IMDB Rating Adjusting for Engagement Over Time



Results

	Top Variables for Full Model	Coefficient Value (Full Model)	Top Variables Pre-1995	Coefficient Value (Pre-1995 Model)	Top Variables Post-1995	Coefficient Value (Post-1995 Model)
1	(Intercept)	8.56331758148543	(Intercept)	8.32481672807817	(Intercept)	9.55362746980077
2	networkCartoon Network	3.60292925818571	caught.daphneTRUE	0.690381455925383	networkThe CW	-3.90211565026293
3	networkBoomerang	3.29420577636898	captured.scoobyTRUE	-0.676424323888735	batmanTRUE	1.4144066471559
4	networkThe WB	2.4191989725819	monster.realTRUE	-0.639014463754104	monster.realTRUE	0.692773762188498
5	scooby.dumTRUE	0.831140543076434	caught.velmaTRUE	-0.511694990427097	networkThe WB	-0.645837013559864
6	networkThe CW	-0.749933324018773	batmanTRUE	-0.49990517703755	caught.notTRUE	0.478661808425985
7	caught.notTRUE	0.704777056003086	snack.shaggyTRUE	0.451889912309716	captured.velmaTRUE	-0.397525096718302
8	scrappy.dooTRUE	-0.479973559759515	scrappy.dooTRUE	-0.450520381161819	groovy	0.337291339948822
9	networkCBS	-0.439805455116329	door.gagTRUE	0.348345681486537	captured.daphneTRUE	0.263379650142656
10	monster.realTRUE	0.400778382551077	snack.velmaTRUE	-0.338240109554917	unmask.scoobyTRUE	0.262282399832238
11	snack.fredTRUE	0.373427809780522	arrestedTRUE	0.335654331095725	networkCartoon Network	0.256594147375614

What Does This Mean?

- Batman had the most significant jump: $-.5$ to 1.4
- Monster.real: changed from $-.64$ to $.69$
- There aren't any more direct variable changes in the series, meaning what people liked and disliked changes throughout the years.
- We can look at character actions!
 - Daphne is well liked. caught.daphne is at $.69$ in the pre-1995 dataset, and captured.daphne is at $.26$ in the post-1995 dataset
 - Velma is not well liked. Caught.velma is at $-.51$ and snack.velma is at $-.33$ in the pre-1995 dataset and captured velma is at $-.4$ in the post-1995 dataset

CONCLUSION

It is possible to predict the IMDB score based on the factors within the episode!

These results also show how culture shifts and different factors are more/less popular over time.

Thank You!

